Simulation v4: Why is the IVB small?

Structural factors that determine IVB magnitude in TWFE panels

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1 Motivation

The v1 simulation showed that $|\text{IVB}|/|\beta|$ is constant in R^2_{within}. This is because the DGP had a single parameter γ_D governing the $D \to Z$ channel, and after TWFE demeaning, $\pi = \gamma_D + \gamma_Y \beta$ regardless of the between/within decomposition of D. The simulation was **tautological**: it only confirmed the formula $\text{IVB} = -\theta^* \times \pi$.

The real question is: **which structural features of a DGP make π and θ^* small after absorbing fixed effects?** This simulation explores four mechanisms.

2 Setup

All mechanisms use the clean case ($\delta = 0$, no confounding through Z), $N = 200$ units, $T = 30$ periods, $\beta = 1$, and 500 replications per scenario. Total: 212 scenarios.

The IVB identity is:

$$\text{IVB} = \hat{\beta}_{\text{long}} - \hat{\beta}_{\text{short}} = -\hat{\theta}^* \times \hat{\pi}$$

where $\hat{\theta}^*$ is the coefficient on Z in the long model and $\hat{\pi}$ is the coefficient on D in the auxiliary regression $Z \sim D + \text{FE}$.

3 Mechanism A: Between/within decomposition of $D \to Z$

3.1 DGP

$$Y_{it} = \beta D_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$
$$Z_{it} = \gamma_D^{\text{btw}} \mu_i^D + \gamma_D^{\text{wth}} (D_{it} - \mu_i^D) + \gamma_Y Y_{it} + \eta_i + \mu_t + \nu_{it}$$

where μ_i^D is the time-invariant unit-level component of D. After TWFE absorbs unit FE, $\gamma_D^{\text{btw}} \mu_i^D$ is removed. The surviving first-stage coefficient is:

$$\pi = \gamma_D^{\text{wth}} + \gamma_Y \beta$$

3.2 Key finding: FE absorbs the between channel

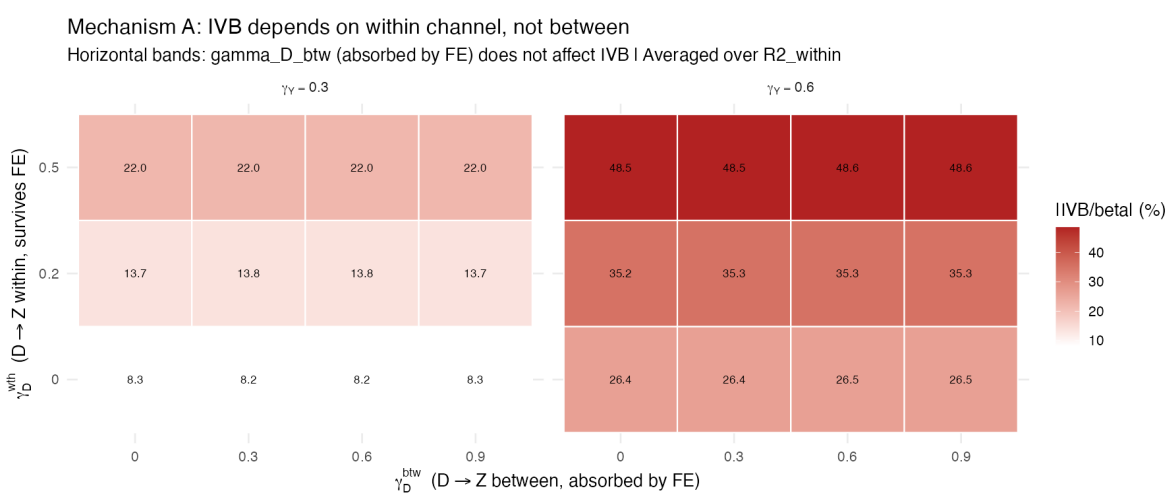

Figure 1: IVB depends on the within channel (γ_D^{wth}, vertical axis), not the between channel (γ_D^{btw}, horizontal axis). Horizontal bands confirm that FE absorbs the between component.

The heatmap shows clear **horizontal bands**: within each row (fixed γ_D^{wth}), the IVB is essentially constant as γ_D^{btw} varies from 0 to 0.9. This confirms that the between channel is fully absorbed by unit FE.

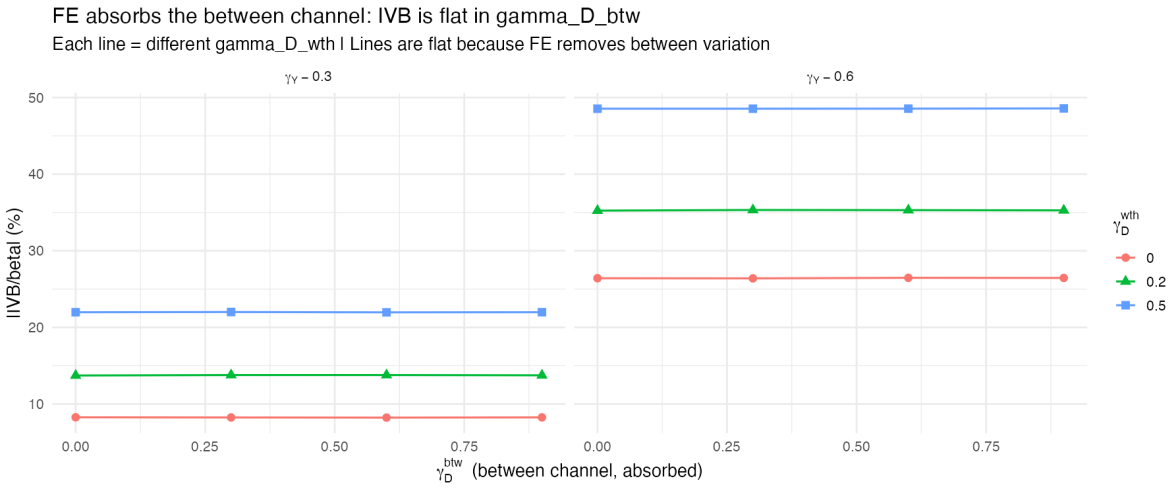

Figure 2: IVB is flat in γ_D^{btw}: the between channel is irrelevant after FE absorption.

3.3 Quantitative verification

Table 1: pi and IVB are invariant to gamma_D_btw (gamma_D_wth=0.2, gamma_Y=0.3).

γ_D^{btw}	$\hat{\pi}$	$\lvert\text{IVB}/\beta\rvert$ (%)
0.0	0.499	13.7
0.3	0.501	13.8
0.6	0.501	13.8
0.9	0.501	13.7

The theoretical prediction is $\pi = 0.2 + 0.3 \times 1.0 = 0.5$, independent of γ_D^{btw}. The simulation confirms this.

3.4 When D causes Z primarily in levels

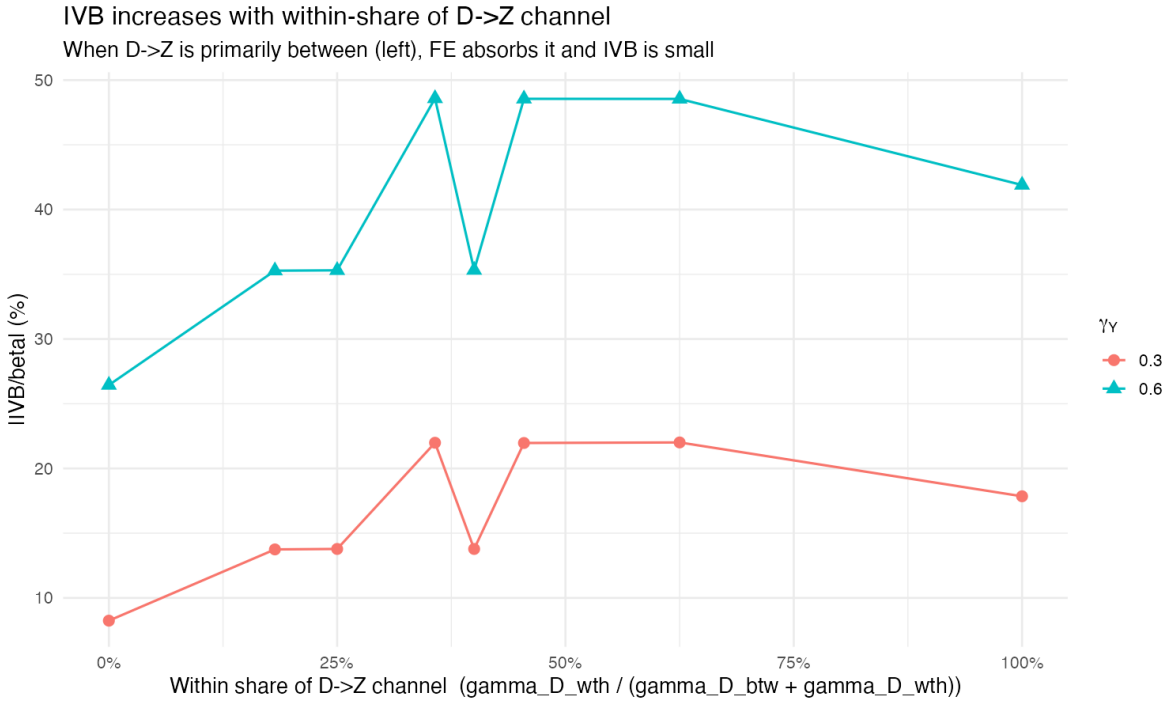

Figure 3: IVB increases with the within-share of the $D \to Z$ channel. When most of the D-Z association is cross-sectional (left side), IVB is small.

Insight for the researcher: Even if D and Z are strongly correlated in the raw data, the IVB can be small if that correlation is cross-sectional (between countries) rather than temporal (within countries), because TWFE absorbs the between channel.

4 Mechanism B: Between/within decomposition of $Y \to Z$

4.1 DGP

$$Z_{it} = \gamma_D D_{it} + \gamma_Y^{\text{btw}} \bar{Y}_i + \gamma_Y^{\text{wth}} (Y_{it} - \bar{Y}_i) + \eta_i + \mu_t + \nu_{it}$$

After TWFE absorbs unit FE, $\gamma_Y^{\text{btw}} \bar{Y}_i$ is removed (since $\bar{Y}_i$ is time-invariant). Therefore θ^* in the long regression depends only on γ_Y^{wth}.

4.2 Key finding: θ^* is independent of $\gamma_Y^{\mathbf{btw}}$

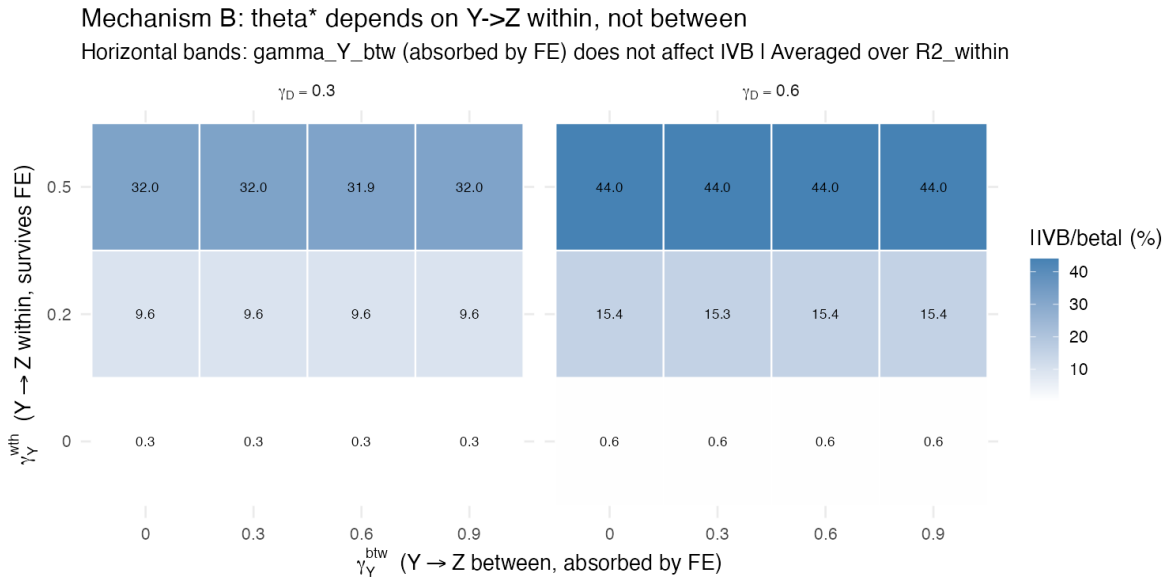

Figure 4: IVB depends on the within channel of $Y \to Z$ (γ_Y^{wth}), not the between channel (γ_Y^{btw}). Analogous horizontal bands.

Table 2: theta* is invariant to gamma_Y_btw (gamma_D=0.3, gamma_Y_wth=0.2).

γ_Y^{btw}	$\hat{\theta}^*$	$\lvert\text{IVB}/\beta\rvert$ (%)
0.0	0.1930	9.6
0.3	0.1919	9.6
0.6	0.1926	9.6
0.9	0.1918	9.6

The range of $\hat{\theta}^*$ across all values of γ_Y^{btw} is only 0.001 — effectively zero.

Insight: If the correlation between Y and Z is primarily cross-sectional (e.g., richer countries have higher inequality in levels), θ^* after FE is small, and IVB is small.

5 Mechanism C: Binary D with staggered adoption

5.1 DGP

$D_{it} = \mathbf{1}(t \geq T_i^*)$ for switchers, with $T_i^* \sim \text{Uniform}\{2, \ldots, T\}$. The fraction of switchers is controlled by `prob_switch`. Never-treated ($D = 0$ always) and always-treated ($D = 1$ always) units contribute zero within-variation.

5.2 Key finding: fewer switchers → larger SE → IVB invisible to inference

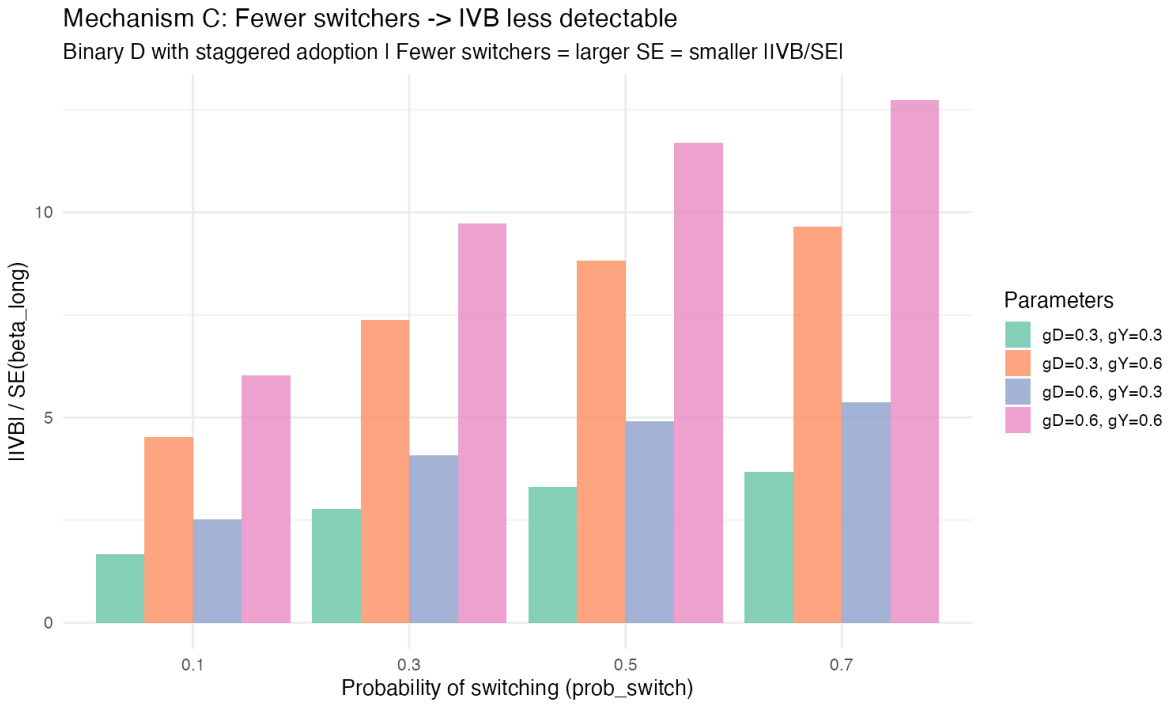

Figure 5: With fewer switchers, |IVB|/SE decreases: the IVB becomes statistically invisible even though it has the same population magnitude.

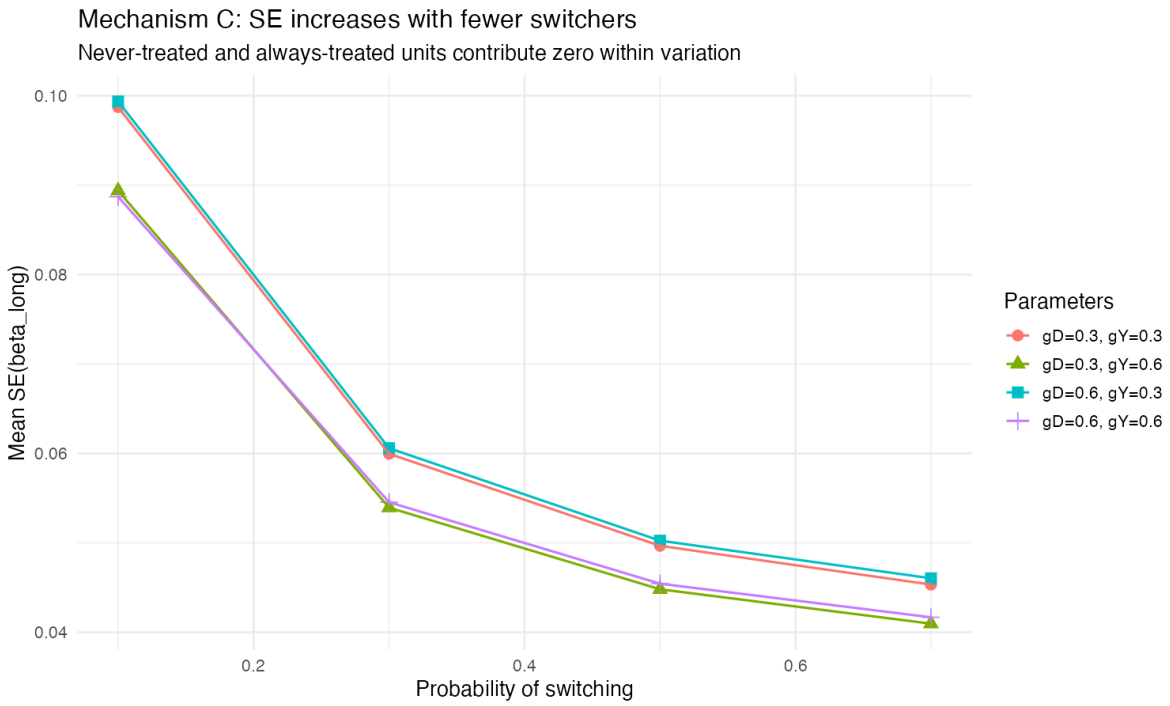

Figure 6: Standard errors increase sharply as the fraction of switchers decreases, because less within-variation in D is available for identification.

Table 3: Population IVB is stable (~27%), but SE grows from 0.045 to 0.099 as switchers decrease.

prob_switch	$\lvert\text{IVB}/\beta\rvert$ (%)	$\lvert\text{IVB}\rvert/\text{SE}$	$\text{SE}(\hat{\beta}_{\text{long}})$
0.1	16.2	1.67	0.0987
0.3	16.6	2.77	0.0600
0.5	16.4	3.31	0.0497
0.7	16.6	3.67	0.0453

Insight: In political science applications where treatment is binary and slow-moving (democratization, peacekeeping deployment), few units actually switch. The IVB exists in population but is **dwarfed by estimation uncertainty**, making it practically irrelevant for inference.

6 Mechanism D: Measurement error in Z

6.1 DGP

The researcher observes $Z_{it}^{\text{obs}} = Z_{it}^{\text{true}} + e_{it}$, where $e_{it} \sim N(0, \sigma_{\text{me}}^2)$. Classic errors-in-variables attenuation bias shrinks θ^* toward zero.

6.2 Key finding: measurement error attenuates θ^* and IVB

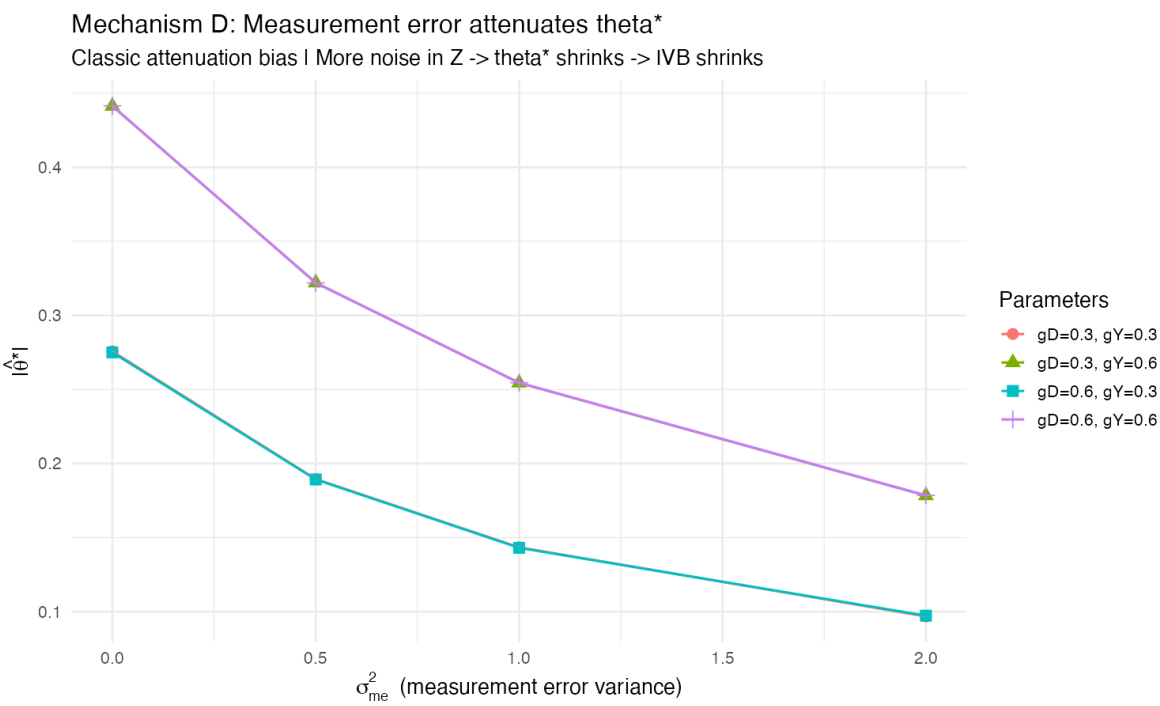

Figure 7: $|\hat{\theta}^*|$ decreases monotonically with measurement error variance, following the classic attenuation bias formula.

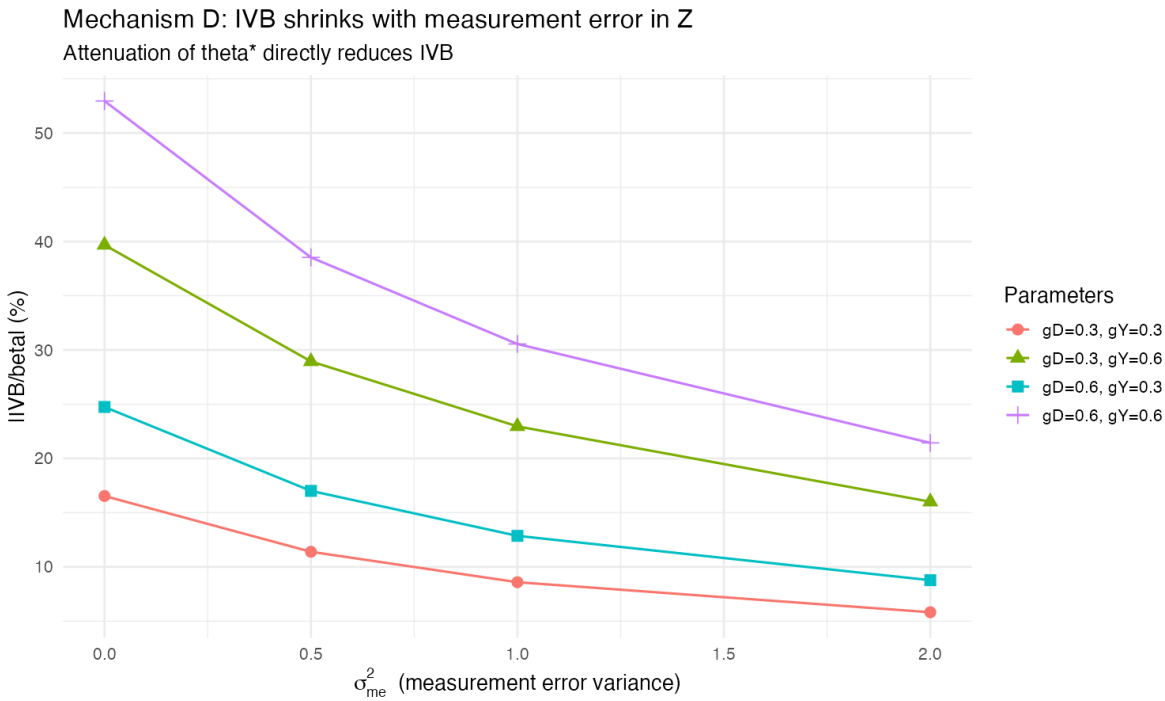

Figure 8: IVB shrinks proportionally with measurement error, because IVB $= -\theta^* \times \pi$ and θ^* is attenuated.

Table 4: theta* is attenuated from 0.28 to 0.10 as measurement error grows from 0 to 2.

σ^2_{me}	$\hat{\theta}^*$	$\lvert\text{IVB}/\beta\rvert$ (%)
0.0	0.2757	16.5
0.5	0.1895	11.4
1.0	0.1434	8.6
2.0	0.0968	5.8

Insight: Socioeconomic control variables in long panels are often measured with substantial error (GDP, inequality indices, governance scores). This measurement error attenuates θ^*, mechanically reducing the IVB.

7 Synthesis

Table 5: Summary across mechanisms. In A and B, one-third of scenarios have IVB smaller than one standard error.

Mechanism	Scenarios	n with $\lvert\text{IVB}\rvert/\text{SE} < 1$	% with $\lvert\text{IVB}\rvert/\text{SE} < 1$	Median $\lvert\text{IVB}/\beta\rvert$ (%)	Max $\lvert\text{IVB}/\beta\rvert$ (%)
A	108	36	33.3	13.75	48.70
B	72	24	33.3	12.48	44.15
C	16	0	0.0	32.15	53.11
D	16	0	0.0	19.22	52.95

7.1 Why is the IVB small in empirical applications?

The empirical studies in the paper (Claassen, Fiscal State, Leipziger, Peacekeeping) share features that activate **all four mechanisms simultaneously**:

1. **Mechanism A** — Treatment variables (D) like democracy or peacekeeping deployment vary primarily across countries (between), not within countries over time. After TWFE absorbs unit FE, the $D \to Z$ channel is weak, so π is small.
2. **Mechanism B** — Outcome variables (Y) like civil war onset or state capacity also correlate with controls (Z) primarily in levels. After FE, θ^* is small.
3. **Mechanism C** — Treatment is binary and slow-moving. Few units switch, so SE is large, and even a moderate IVB is dwarfed by estimation uncertainty.
4. **Mechanism D** — Control variables like GDP, ethnic fractionalization, or governance indices are measured with error, especially in long panels. This attenuates θ^*.

These mechanisms are **mutually reinforcing**: a study can have small IVB because of any one of them, and empirical studies typically exhibit all four.

7.2 Robustness: coverage rates

Table 6: 95% CI coverage of the short model (should be ~0.95). All mechanisms are well-calibrated.

Mechanism	Mean	Min	Max
A	0.951	0.926	0.978
B	0.951	0.916	0.968
C	0.949	0.930	0.962
D	0.949	0.928	0.968

8 Suggested next steps

1. **Combine mechanisms A+C**: A DGP with binary D *and* between/within decomposition of $D \to Z$. This is the most realistic scenario for political science and would show even smaller IVB than either mechanism alone.
2. **Calibrate to empirical studies**: Extract R^2_{within}, proportion of switchers, and approximate measurement error from the four replication studies. Run the simulation with those calibrated parameter values to produce study-specific IVB bounds.
3. **Formal bound**: Derive an analytical upper bound for |IVB|/SE as a function of observable quantities (within-share of D, first-stage R^2, number of switchers). This would give researchers a formula to assess IVB risk without running simulations.
4. **Integration with v3 diagnostic tool**: The `ivb_diagnostic()` function (v3 plan, Part 1) estimates $\hat{\pi}$ and $\hat{\theta}^*$ from actual data. Combining it with the structural insights from v4 would let researchers say: "My estimated |IVB|/SE is small, and here are the structural reasons why."
5. **Paper section**: Write up Mechanisms A and B as the main theoretical contribution (formal result: FE absorbs between channels). Mechanisms C and D are supporting arguments that reinforce the conclusion.

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